

Unit 9, Lesson 7a: Concepts of Density Based on Volume



Benchmark

MA.912.GR.4.5 Solve mathematical and real-world problems involving the volume of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.



Learning Target

- ☐ I can solve mathematical and real-world problems involving concepts of density based on volume.

Teacher Notes: This lesson is an excellent opportunity for mathematics and science teachers to collaborate. Similar concepts are taught in grade 8 and Physical Science. Teachers are encouraged to discuss when and how density and mass are taught in science. This collaboration can benefit a student's learning experience.

The warm-up is designed to determine any weaknesses in students' prior knowledge of density. As you circulate, listen for common misconceptions.



9.7a.1 Warm-Up: Considering Density

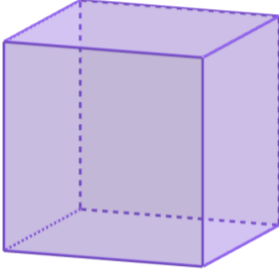
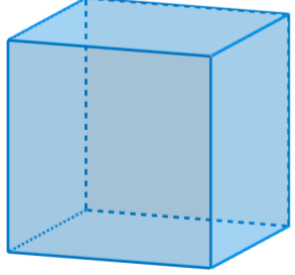
Suppose you have two boxes with the same dimensions. In one box are 5 cotton balls and in the other box are 5 quarters.

1. Sketch a picture of the boxes.
Answers will vary.

2. Which box would likely be fuller than the other?
Cotton Balls

3. Which box would likely be heavier than the other?
Quarters

4. Explain whether it is possible to add more cotton balls or more quarters so that each box has the same weight.
Answers will vary. Allow students to have some productive struggle and/or creative responses.

Cotton Balls	Quarters
	

5. Anthony exchanged the box of 5 quarters for a box of 25 cotton balls. He described the box of 25 cotton balls as being more dense than the box of 5 cotton balls. Explain why you think Anthony used the word “dense”?

Answers will vary.

Teacher Notes: For the purpose of the lesson, teachers may want to begin the Guided Instruction by polling students on their response to this question. Teachers may wish to use the responses to comprise an agreed upon definition of density. Alternatively, teachers may wish to ask the science department for the definition used in science classes.



9.7a.2 Collaboration: Density, Mass, and Weight

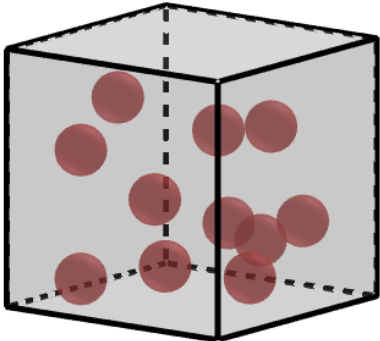
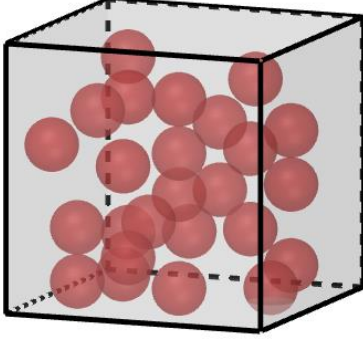
Teacher Notes: This collaboration is designed to help students solidify the concept of density as well as to model the proper use of mass and weight. Teachers may wish to discuss with the science department the use of these terms. Although on Earth, the terms are often interchanged, they do not mean the same thing. Model for your students the use of correct language.

As you circulate, listen for possible misconceptions, incorrect interchange of mass and weight, and for prior knowledge that may need support during the Guided Instruction.

Density, mass, and weight are related to each other but have very different meanings and applications. Often mass and weight are confused with each other, but they are not the same.

Anthony described the box of 25 cotton balls as being denser than the box of 5 cotton balls. This is because the volume of the box of 25 cotton balls was fuller than the volume of the box of 5 cotton balls.

Density can have many meanings and applications. In the context of volume, density indicates the degree to which the space is filled or occupied.

Low Density	High Density
	

Different materials have different density because the atoms or molecules that make up the material can occupy the space differently.

For example, water has a density of 1 gram per cubic centimeter ($\frac{\text{g}}{\text{cm}^3}$), which means that 1 gram (g) of water will fit into a box that is 1 centimeter (cm) by 1 cm by 1 cm. Objects that float have a density that is between $0 \frac{\text{g}}{\text{cm}^3}$ and $1 \frac{\text{g}}{\text{cm}^3}$. Objects that sink have a density that is greater than $1 \frac{\text{g}}{\text{cm}^3}$.

1. Compare the density of each object with the density of water. Then, select the range of values that most likely represents the object's density.

Object	Density Between $0 \frac{\text{g}}{\text{cm}^3}$ and $1 \frac{\text{g}}{\text{cm}^3}$	Density Greater Than $1 \frac{\text{g}}{\text{cm}^3}$
Rock	☐	☐
Quarter	☐	☐
Ping Pong Ball	☐	☐
Empty Plastic Bottle	☐	☐

Teacher Notes: Students who struggle with this question may need further support on the concept of density. A glass of water, a coin, and an object that floats may help students visualize.

All objects have a mass and a weight. Weight changes with the gravitational pull on an object. On Earth, that gravitational pull is consistent, so the weight of an object in Florida is the same as the weight of the object in Iceland. The official unit for weight is Newtons, but this unit is rarely used when describing common objects. Instead, the units used for mass—such as grams, pounds, or tons—are typically used as the units for weight.

If an object weighs 100 pounds (lb.) on Earth, it will weigh 37.7 lb. on Mars and 236.4 lb. on Jupiter. This is because the gravitational pull varies on each planet. The same object would have a weight of 0 lb. in space. Yet, the mass of the object will be 100 lb. on Earth, Mars, Jupiter, and space because mass does not change with the gravity.

2. An object on Earth that has a mass of 10 kilograms (kg) would have a weight of 98 Newtons. An object on the Moon that has a mass of 10 kg would have a weight of 16 Newtons.

Discuss with your partner why an object with the same mass has a different weight in Newtons on Earth and on the Moon. Summarize your discussion.

Answers will vary. Students who struggle with this may need reading support or to have a verbal explanation.



9.7a.3 Guided Instruction: Relating Density, Mass, and Volume

The mass of an object, the object's density, and the space the object occupies, also known as volume, are related. The formula is commonly written as shown.

$$\text{Density} = \frac{\text{mass}}{\text{volume}} \text{ or } D = \frac{M}{V}$$

1. In the Collaboration, it was stated that water has a density of $1 \frac{\text{g}}{\text{cm}^3}$.

- The numerator of $\frac{\text{g}}{\text{cm}^3}$ is a unit of mass.
- The denominator of $\frac{\text{g}}{\text{cm}^3}$ is a unit of volume.

2. Rewrite $D = \frac{M}{V}$ in terms of each variable.

a. $M = D \cdot V$

b. $V = \frac{M}{D}$

Teachers are encouraged to allow students to use the equivalent forms of the formula to solve problems algebraically. The lesson builds the complexity of using the formula for different unknowns. Teachers may need to return some students back to the equivalent forms.

3. Discuss with your partner whether the volume would have to change if the mass of the water was 1 lb. Summarize your discussion.

Answers will vary. Students often do not have a conceptual knowledge of units. Teachers may wish to prepare to have a common item that has a measurement in g and a common item that has a measurement in lb. Teachers may also wish to have a box that is 1 cm^3 and a water bottle that is 16 oz. A full water bottle of 16 oz. weighs 1 lb.

4. Discuss with your partner what makes the statement shown incorrect. Summarize your discussion.

Water weighs 62.4 pounds per cubic foot.

Answers will vary. The lesson has laid the groundwork for students to understand that mass and weight are not the same. This question is designed to be the exclamation point to that difference. Help students edit the statement so that it helps to solidify that weight and mass are not the same. Suggested edit: Water has a mass of 62.4 lb. per cubic foot.

5. Limestone is often used to create sculptures. An artist has a block of limestone that has the dimensions of 15 cm by 15 cm by 15 cm. The density of the limestone is $2.688 \frac{\text{g}}{\text{cm}^3}$. Determine the mass, in g, of the limestone.

9,072 g

Teachers may wish to ensure student understanding of the units of the answer by including the units in their work.

6. Determine the density of a block of limestone that has a mass of 20 lb. and has the dimensions of 5.91 inches (in.) by 5.91 in. by 5.91 in.

$0.097 \frac{\text{lb.}}{\text{in.}^3}$

Teachers may wish to ensure student understanding of the units of the answer. Teachers should determine what place value to ask students to use. Science teachers often use significant digits, so teachers may wish to reflect that when a specific rounding value is not stated. Other teachers may wish to always round to the nearest thousandth.

7. A block of limestone with dimensions of 15 cm by 15 cm by 15 cm is the same size as a block of limestone with dimensions 5.91 in by 5.91 in. by 5.91 in.

Discuss with your partner why the density for the block of limestone with dimensions of 5.91 in. by 5.91 in. by 5.91 in. is not the same as the density given for the block of limestone with the dimensions of 15 cm by 15 cm by 15 cm. Summarize your discussion.

Answers will vary. Teachers may use this question to consider whether students have any misconceptions on units for mass or density.

In science journals the density of an object it is often stated using the units $\frac{\text{g}}{\text{cm}^3}$, but the units may vary when the mass or the volume of the object is larger. If needed, scientists can easily convert the units.

8. Select all of the units that could represent density.

☐ $\frac{\text{ft}}{\text{sec}}$

☒ $\frac{\text{kg}}{\text{m}^3}$

☐ $\frac{\text{mL}}{\text{in.}^2}$

☒ $\frac{\text{oz}}{\text{in.}^3}$

☐ $\frac{\text{mm}^3}{\text{g}}$

Teachers may wish to refer back to the question about a pound of water fitting in a cube that has a volume of 1 cubic centimeter.

9. The density of a silver bar is $0.321 \frac{\text{lb.}}{\text{in.}^3}$. The dimensions of a silver bar is 5 in. by 12 in. by 3.25 in. Determine the mass, in lb., of the bar. If necessary, round to the nearest thousandth of a lb.

62.595 lb.

Teachers may wish to show students the work and include the units to ensure students understand that the units follow the same rules in an equation as values.

$$0.321 \frac{\text{lb.}}{\text{in.}^3} = \frac{M}{5 \text{ in.} \cdot 12 \text{ in.} \cdot 3.25 \text{ in.}}$$

$$0.321 \frac{\text{lb.}}{\text{in.}^3} = \frac{M}{195 \text{ in.}^3}$$

$$0.321 \frac{\text{lb.}}{\text{in.}^3} \cdot 195 \text{ in.}^3 = \frac{M}{195 \text{ in.}^3} \cdot 195 \text{ in.}^3$$

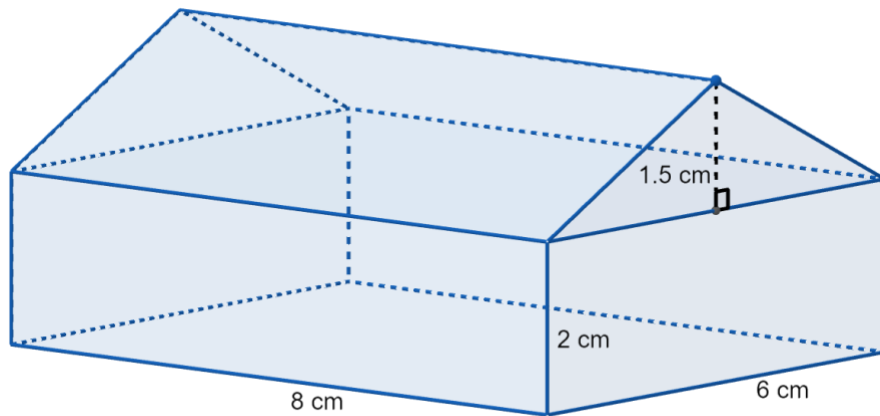
10. A gold bar has dimensions of 4 in. by 11 in. by 2 in. The mass of the gold bar is 61.336 lb. Determine the density, in $\frac{\text{lb.}}{\text{in.}^3}$, of the gold bar.

$$0.697 \frac{\text{lb.}}{\text{in.}^3}$$



9.7a.4 Guided Practice: Determining Density, Mass, and Volume

1. A product designer is creating a new container for maple syrup, as shown.



The density of maple syrup is $1.37 \frac{\text{g}}{\text{cm}^3}$. Determine the mass, in g, of the container.
180.84 g



9.7a.5 Your Turn!

1. Sebastian has a workout ball that is filled with sand. The density of sand is 101.82 pounds per cubic foot ($\frac{\text{lb.}}{\text{ft.}^3}$). The diameter of the ball is 1 foot (ft.). Determine the mass, to the nearest lb., of the workout ball.
53 lb. (Note: This answer is true no matter which value of π a student chooses.)

2. The densities, in $\frac{\text{g}}{\text{cm}^3}$, of three different materials are shown.

High Density Polyethylene (Plastic)	Aluminum	Oak
0.95	2.7	0.82

For each material, determine the length of one side of a cube that has a mass of 100 grams (g). Round to the nearest hundredth of a centimeter (cm).

	High Density Polyethylene (Plastic)	Aluminum	Oak
Side Length	4.72 cm	3.33 cm	4.96 cm



9.7a.6 Wrap-Up: Ticket Out the Door

1. A propane tank is in the shape of a cylinder. The propane tank has a mass of 290 lb. when it is empty. The diameter of the tank is 2.5 ft. and the height of the tank is 3.8 ft. The density of propane is $36.28 \frac{\text{lb.}}{\text{ft.}^3}$.

Determine the weight, in lb., of the tank when it is filled with propane. Round to the nearest tenth of a lb.

Answer will vary depending on the value of π used.

Value of pi	Answer
π	676.74
3.14	676.4
3.1415	676.72
$\frac{22}{7}$	677.01